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Software Engineering(24CS209P)

Lab Experiment - 1

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Use case:

- Issue Book

Scope:

- The scope of this use case is the **Library Management System**, which allows librarians to issue books to registered users.

Description:

- This use case describes how a librarian issues a book to a student using the library system.

Flow Description:

- This use case explains the interaction between the **librarian** and the **system** to successfully issue a book to a student.

Precondition:

- The librarian is logged into the system.
- The student is a registered library member.
- The requested book exists in the library database.

Activation:

“This use case starts when the librarian selects the **Issue Book** option from the system menu.”

Main flow:

1. Librarian enters the student ID in the system.
2. System validates the student's account.
3. Librarian enters or scans the book ID.
4. System checks the availability of the book.
5. System assigns the book to the student.
6. System updates the book status as **Issued**.
7. System displays the due date for the book.

Alternate Flow:

(A1) Book Not Available

1. System displays the message "Book not available for issue."

(A2) Student Reached Issue Limit

1. System displays the message "Maximum book issue limit reached."

Exceptional flow:

- System fails due to database or network error; an error message is displayed.

Termination:

- The system returns to the librarian dashboard.

Post condition:

- The book is successfully issued to the student.
- The database is updated with issue details and due date.